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Biomonitoring of Perfluoroalkyl Acids in Human Urine and Estimates of Biological Half-Life

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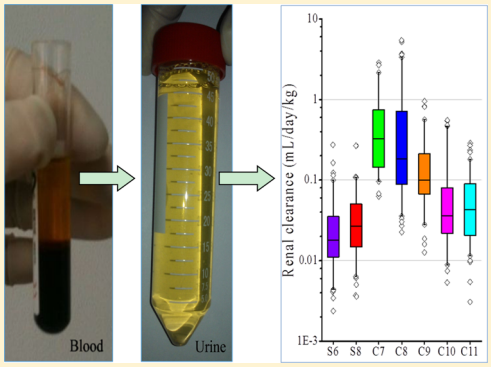
Supporting Information

ABSTRACT: Perfluoroalkyl acids (PFAAs) are persistent and bioaccumulative compounds that have been associated with adverse health outcomes. In human blood, PFAAs exist as both linear and branched isomers, yet for most linear homologues, and for all branched isomers, elimination rates are unknown. Paired blood and urine samples (n = 86) were collected from adults in China. They were analyzed by a sensitive isomer-specific method that permitted the detection of many PFAAs in human urine for the first time. For all PFAAs except perfluoroundecanoate (PFUnA), levels in urine correlated positively with levels in blood. Perfluoroalkyl carboxylates (PFCAs) were excreted more efficiently than perfluoroalkane sulfonates (PFSAs) of the same carbon chain-length. In general, shorter PFCAs were excreted more efficiently than longer ones, but for PFSAs, perfluorooctanesulfonate (PFOS, a C8 compound) was excreted more efficiently than perfluorohexanesulfonate (PFHxS, a C6 compound). Among PFOS and perfluorooctanoate (PFOA) isomers, major branched isomers were more efficiently excreted than the corresponding linear isomer. A one-compartment model was used to estimate the biological elimination half-lives of PFAAs. Among all PFAAs, the estimated arithmetic mean elimination half-lives ranged from 0.5 ± 0.1 years (for one branched PFOA isomer, 5m-PFOA) to 90 ± 11 years (for one branched PFOS isomer, 1m-PFOS). Urinary excretion was the major elimination route for short PFCAs (C ≤ 8), but for longer PFCAs, PFOS and PFHxS, other routes of excretion likely contribute to overall elimination. Urinary concentrations are good biomarkers of the internal dose, and this less invasive strategy can therefore be used in future epidemiological and biomonitoring studies. The very long half-lives of long-chain PFCAs, PFHxS, and PFOS isomers in humans stress the importance of global and domestic exposure mitigation strategies.



INTRODUCTION

After decades of production and applications, perfluoroalkyl acids (PFAAs) are ubiquitous in the environment and have been detected in humans all around the world.[1,2] The most commonly detected PFAAs in human samples are perfluorooctanesulfonate (PFOS), perfluorooctanoate (PFOA), and perfluorohexanesulfonate (PFHxS). Studies have shown that exposure to PFOS and PFOA can be associated with adverse effects including lower birth weight,[3] hyperuricemia,[4] chronic kidney disease,[5] and immunotoxicity.[6] Fewer studies have examined adverse health effects of PFHxS, although recently it has been associated with attention deficit hyperactivity disorder in children.[7,8]

The rate at which a chemical is eliminated from human body (i.e., the biological half-life, or clearance) is an important consideration in its respective hazard profile. A toxicant with a longer elimination half-life generally indicates a greater potential to bioaccumulate following repeated exposures, or that it will take longer to be eliminated from the body after exposure cessation. Thus, elimination half-lives data are important pharmacokinetic parameters for many PFAAs which are still undergoing regulatory scrutiny, or which are already regulated in some jurisdictions.

The arithmetic mean serum elimination half-lives of PFOS, PFOA, and PFHxS in humans were estimated to be 5.4, 3.8, and 8.5 years, respectively, in a highly exposed occupational cohort of 24 males and 2 females.[9] Likewise, the serum elimination half-life of PFOA in residents exposed through a contaminated public water supply was estimated to be 2.3 or 2.9 years.[10,11] However, there are currently no human clearance data for most other PFAAs, including various other perfluoroalkyl carboxylates (PFCAs, i.e., C7 and C9−C11) and some PFOS-precursors, such as perfluorooctanesulfonamide (PFOSA). Furthermore, although data are available on PFOS and PFOA elimination half-lives, these studies did not discriminate between the linear and various branched isomers, known to be present in human blood,[12−15] and which displayed

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a wide range of pharmacokinetic behaviors in animal studies.[16−19]

The isomers of PFOA and PFOS result from the electrochemical fluorination (ECF) manufacturing method. ECF was used to produce PFOA beginning in 1947 and PFOS and its precursors beginning in 1949.[20,21] This process was known to yield a complex, yet rather consistent, mixture of linear (ca. 70% for PFOS, 80% for PFOA) and branched isomers (ca. 30% for PFOS, 20% for PFOA) in final products.[22] Curiously, for PFOS, all studies on the isomer compositions in human blood have consistently shown that the proportion of total branched PFOS isomers is higher than that (∼30%) in commercial PFOS mixtures.[13,14] This apparent preferential bioaccumulation of branched PFOS isomers in humans is opposite to what is anticipated from PFOS isomer pharmacokinetic studies in rodents,[16,17] whereby PFOS branched isomers are excreted more efficiently in urine than their linear counterparts; thus a study on human renal clearance of PFOS isomers is required before sources of human contamination will be fully understood.

Previous animal studies have demonstrated that urine was the primary elimination route for PFOA and PFOS in rats and for PFOA in monkeys,[23,24] and fecal elimination became increasingly important for PFCAs with longer carbon chain length (C > 8) in rats,[25] whereas in humans the main mode(s) of elimination have not yet been confirmed, although recently Beesoon et al.[26] reported urine as one mode of PFHxS excretion in a highly exposed Canadian family. No other studies have compared excretion pathways in the same participants, and few studies have even reported PFAA concentrations in urine even though it was known as one of the excretion modes for humans in 1980.[27] In a study of 20 participants,[28] renal clearance rates of PFOA and PFOS were estimated, however, the isomers of PFOS and PFOA were not examined separately, and no other PFAAs were examined. Other elimination routes, including hair and nail growth are likely minor,[29,30] whereas in premenopausal women menstrual bleeding,[28] pregnancy,[31] and lactation[2] are likely important excretion modes.

Here we developed a highly sensitive analytical method for urine and analyzed the concentrations of various PFAAs, including individual isomers of PFOS and PFOA, in 86 paired Chinese human blood and urine samples for the first time. Correlations between urine and blood levels were examined, while rates of renal clearance and half-lives were estimated and were discussed with respect to human exposure.

■ MATERIALS AND METHODS

Nomenclature and Acronyms. The acronyms of all PFAAs and PFOSA used here are listed in Table S1 of the Supporting Information (SI). For specific isomers of PFOA and PFOS, previous nomenclature was adopted.[32] Using PFOS as an example, linear and perfluoroisopropyl branches are abbreviated as *n*- and *iso*-PFOS, respectively. For the remaining monoperfluoromethyl isomers, *m* refers to the perfluoromethyl branch and the preceding number indicates the carbon position of the branching point (e.g., 3*m*-PFOS, 5*m*-PFOS). For total dimethyl-substituted-branched isomers, these could not be distinguished and are labeled as $\sum m_2$-PFOS. Branched isomers of all other perfluoroalkyl sulfonates (PFSAs) and PFCAs were rarely detected, thus reported concentrations of PFHxS, perfluoroheptanoate (PFHpA), perfluorononanoate (PFNA), perfluorodecanoate (PFDA), and perfluoroundecanoate (PFUnA) are exclusively for the linear isomer.

PFAA and PFOSA Standards. The 3M Co. donated authentic ECF PFOS [ca. 70% linear and 30% branched isomers, by ^{19}F nuclear magnetic resonance (NMR)] and PFOA (ca. 78% linear and 22% branched, by ^{19}F NMR) standards. All other PFAA and PFOSA standards, including brPFOSK, TPFOA, PFAC-MXB, MPFAC-MXA, PFOSA, and M8FOSA-M were purchased from Wellington Laboratories (Guelph, ON, Canada). The relative percentages of linear and branched components, as determined by ^{19}F NMR, in brPFOSK (78.8% linear, 10% *iso*-PFOS, 1.2% 1*m*-PFOS, 1.9% 3*m*-PFOS, 2.2% 4*m*-PFOS, 4.5% 5*m*-PFOS, 0.71% $\sum m_2$-PFOS) and TPFOA (79% linear, 9% *iso*-PFOA, 3% 3*m*-PFOA, 4% 4*m*-PFOA, 4.5% 5*m*-PFOA) were provided by Wellington Laboratories. PFAC-MXB is a mixture of linear standards, including PFHpA, PFOA, PFNA, PFDA, PFUnA, PFHxS, and PFOS. MPFAC-MXA is a mixture of linear mass labeled internal standards for those PFAAs except PFHpA, and the internal standard of PFOA was used for the quantification of PFHpA. M8FOSA-M is a linear mass labeled internal standard for PFOSA. No standards were available for branched PFOSA isomers.

Sample Collection. The College of Environmental Science and Engineering, Nankai University, China, and the Health Research Ethics Board, University of Alberta, Canada, approved this research. Eighty-six paired blood (56 serum and 30 whole blood) and morning urine samples were collected from healthy volunteers in Shijiazhuang and Handan, Hebei province, China, in April−May 2010. Shijiazhuang is the capital city, while Handan is an industrial city, both with estimated populations of 10 million. Demographic information of the participants is given in SI Table S2.

Sample Extraction. Solid-phase extraction cartridges, Oasis-HLB (Waters, 150 mg/6 mL) and Oasis-WAX (Waters, 150 mg/6 mL), were used for serum and urine samples, respectively. Briefly, 0.5 mL of serum or 50 mL of urine was spiked with mass labeled internal standards (1 ng of each) and the mixture was thoroughly mixed on a vortex. The cartridges were conditioned with 2 mL of methanol (high-performance liquid chromatography (HPLC) grade) followed by 2 mL of 0.1 M formic acid. The serum or urine was then loaded to the column and washed successively with 3 mL of 0.1 M formic acid, 5 mL of 50% 0.1 M formic acid/50% methanol, and 1 mL of 1% ammonium hydroxide. The cartridge was dried by vacuum and PFAAs and PFOSA were eluted with 3.0 mL of 1% ammonium hydroxide in methanol. The eluate was evaporated to near-dryness under a gentle stream of high-purity nitrogen at 55 °C and then reconstituted with a mixture of 300 μL of methanol and 200 μL of HPLC grade water. The extract was transferred to a polypropylene tube and centrifuged at 12 000 rpm for 20 min. Finally, the supernatant was transferred to an autosampler vial for HPLC-tandem mass spectrometry (MS/MS) analysis.

The ion-pairing method of Hansen et al.[33] was used with minor modifications to extract PFAAs and PFOSA from the 30 whole blood samples. In brief, 1 ng of mass labeled internal standards was spiked in 0.5 mL of blood in a 15-mL polypropylene tube, followed by addition of 1 mL of 0.5 M tetrabutylammonium hydrogensulfate (TBAH, pH 10), and 2 mL of 0.25 M sodium carbonate buffer. After mixing on a vortex, 5 mL of methyl tert-butyl ether (MTBE) was added, and the mixture was shaken for 20 min. The organic and the aqueous phases were separated by centrifugation at 3000 rpm for 10 min. Four mL of MTBE was removed and transferred to

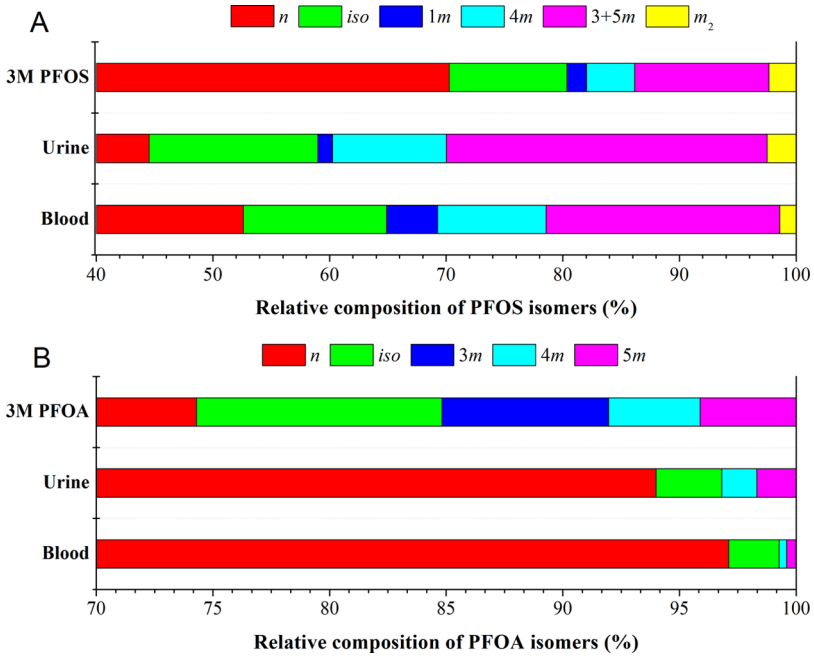

Figure 1. Isomer compositions of (A) PFOS and (B) PFOA in the paired blood−urine samples.

a new 15-mL polypropylene tube. The extraction procedure was repeated two more times as described above. All the organic phases were combined and the solvent was evaporated to near-dryness under gentle nitrogen flow, then reconstituted with a mixture of 300 μL of methanol and 200 μL of pure water. The mixture was centrifuged at 12 000 rpm for 20 min at 4 °C, and 400 μL of supernatant was collected for HPLC-MS/MS analysis.

Urinary creatinine concentrations were measured using a picric acid method. The detailed descriptions of urinary creatinine analysis, and the HPLC-MS/MS method for PFAAs and PFOSA analysis are provided in the SI. The chromatograms of PFOS, PFOA, and PFOSA isomers are shown in SI Figure S1.

Quality Control. A method blank (HPLC grade water) was extracted with each batch of 10 samples to check background contamination, and one solvent blank (HPLC grade methanol) was injected after every five samples to monitor any instrument carryover. Triplicate recovery experiments were performed with calf serum or blood, and with synthetic urine at three levels (2, 20, and 200 ng/L). The recoveries of all 16 PFAAs (including isomers) and linear PFOSA, and the method limit of detection (LOD) were satisfactory (see SI Table S3). The calf serum and blood, as well as the synthetic urine, were purchased from Lampire Biological Laboratories (Pipersville, PA, USA). The results indicate that 3M ECF PFOS and PFOA at 10 ng/mL constituted 70.26 and 74.29% linear isomers (Figure 1), which were very consistent with the reports of Kestner (3M ECF PFOS, 70% linear) by ^{19}F NMR and Stevenson (3M ECF PFOA, 10 ng/mL, 74.7% linear) by LC-MS/MS.[34,35]

Renal Clearance Estimation. Renal clearance can be defined as the volume of serum from which a chemical is completely removed in a given time period. Daily renal clearance of individual PFAAs was calculated based on the paired serum and urinary concentrations (eq 1).

$$\mathrm{CL}_{\mathrm{renal}} = \frac{C_{\mathrm{urine}} \times V}{C_{\mathrm{blood}} \times W} \tag{1}$$

$\mathrm{CL}_{\mathrm{renal}}$ is renal clearance (mL/day/kg), C_{urine} is the concentration of individual PFAA or PFOSA in urine (ng/L or ng/g creatinine), and V is daily urine excretion volume (L/day) or daily creatinine excretion value (g/day). Normal females and males excrete about 1.2 and 1.4 L urine every day,[36] and excrete about 0.9 and 1.1 g creatinine every day.[37] C_{blood} is individual serum concentration (ng/mL). W is body weight (kg); the body weights of females and males are estimated as 55 and 65 kg, respectively.

Estimation of Half-lives of PFAAs. A one-compartment model (eq 2) was used to estimate the biological elimination half-lives of PFAAs.

$$T_{1/2} = \frac{0.693 \times V}{\mathrm{CL}_{\mathrm{total}}} \tag{2}$$

$T_{1/2}$ is biological half-life (days), $\mathrm{CL}_{\mathrm{total}}$ is total clearance (mL/day/kg), and V (mL/kg bw) is the volume of distribution, defined as the total amount of a substance in the body divided by its concentration in serum. Thompson et al.[38] reported that V was 170 and 230 mL/kg for PFOA and PFOS in humans, while Ohmori et al.[39] reported that V was not much different between PFCAs, or between sexes in rats, although the V value of PFDA was larger than those of other PFCAs. In addition, Han et al.[40] summarized the V values for different mammals and found that the V of PFOA was quite consistent among the mammals with a mean value of 191 ± 67 mL/kg. Thus, to simplify the estimation, the V values of 170 and 230 mL/kg were used to estimate the half-lives for all PFCAs and PFSAs, respectively.

Statistical Analysis. For the purpose of descriptive statistics, concentrations below the LOD were substituted by LOD divided by the square root of 2.[41] All the data were log transformed before applying statistical tests. ANOVA, followed by Bonferroni's test was used to assess differences in the PFAA concentrations in blood and matched urine samples among gender and age groups. Paired t tests were used to assess the difference in isomer proportions in blood and urine samples, the difference in renal clearance between urinary PFAA

concentrations normalized by volume or creatinine, and the difference in renal clearance between different PFAAs. To assess any relationship between PFAA concentrations in blood and the matched urine samples, we calculated Spearman rank-order correlations. IBM SPSS Statistics version 20 (Chicago, IL) was used to analyze these data, and significance was set to $p < 0.05$.

RESULTS

Total PFAA Concentrations in Blood and Urine. For the 30 participants for whom whole blood was analyzed, serum concentrations were estimated by accounting for the average hematocrit content of whole blood in the Chinese population: 41.0% for females and 45.8% for males.[42] Thus, whole blood PFAA concentrations were divided by a factor of 0.590 or 0.542 to give a reliable estimate of the serum concentrations of females and males, respectively. The major PFAAs detected in the blood samples, reported here as median concentrations, were total PFOS (∑PFOS, 19 ng/mL), ∑PFOA (2.3 ng/mL), PFHxS (1.2 ng/mL), PFNA (0.37 ng/mL), PFUnA (0.18 ng/mL), and PFDA (0.15 ng/mL). Descriptive statistics on other PFAAs and PFOSA detected in the blood samples are given in Table 1. It is notable that linear PFOSA was detected in all whole blood samples (median 0.029 ng/mL), but was not detected in any serum samples.

To examine if PFAA blood concentrations were affected by age or gender, the 86 participants were divided into four categories, namely, young females (age ≤50 years, $N = 20$), older females (> 50 years, $N = 19$), young males (≤ 50 years, $N = 32$), and older males (> 50 years, $N = 15$) (see SI Figure S2). The mean concentrations of PFHxS ($p < 0.001$), ∑PFOS ($p < 0.05$), and PFNA ($p < 0.05$) in the young female group were significantly lower than in the other three groups. No significant difference was observed among the other three groups for all PFAAs. Thus, the older female group and male groups were combined for subsequent discussion.

The concentrations of PFAAs in urine were normalized to both urinary volume (ng PFAA/L urine) and creatinine content (ng PFAA/g urinary creatinine) (see Table 1). Strong correlations ($p < 0.001$, $r = 0.517–0.932$) were observed between the concentrations of each PFAA (including individual isomers) expressed with either units (see SI Table S4). The four major PFAAs in urine were ∑PFOS and ∑PFOA, followed by PFNA and PFHxS. For most PFAAs, a very good correlation (i.e., $p < 0.05$) was observed between the blood and urine concentrations (see SI Table S4). The weakest correlations were observed for PFUnA ($p > 0.05$), which is the longest chain-length PFCA that could be routinely detected in urine (detection frequency 98%), and for the branched isomer 3*m*-PFOA, which was only detected in 23% of samples (versus >75% for all other PFOA isomers). Among all urine samples, the predominant PFAAs (median values were reported and normalized by urinary volume) were PFOS (25 ng/L), followed by PFOA (19 ng/L), PFNA (1.7 ng/L), PFHxS (1.1 ng/L), PFHpA (0.82 ng/L), PFUnA (0.30 ng/L), and PFDA (0.22 ng/L). Linear PFOSA was detected in most urine samples with median concentration of 0.44 ng/L (see Table 1).

Isomer-Specific Analysis of PFOS and PFOA. Figure 1 shows the isomer compositions of PFOS and PFOA in human samples compared to authentic 3M ECF manufactured standards. For PFOS, the major isomer in the blood samples was *n*-PFOS, with a mean percentage of 53%, which is significantly lower than in the ECF standard (70%). The other

Table 1. Summary Statistics of PFAAs and PFOSA Concentrations in Human Serum ($n = 56$), Whole Blood ($n = 30$), and Matched Urine Samples ($n = 86$)[a]

		PFHxS	PFOS							PFHpA	PFNA	PFDA	PFUnA	PFOA								linear PFOSA
			n	*iso*	1*m*	4*m*	3+5*m*	m_2	total					*n*	*iso*	3*m*	4*m*	5*m*	total	
blood ($n = 86$) ng/mL	mean	2.6	15	4.0	1.7	3.5	7.2	0.49	31	0.085	0.45	0.19	0.21	3.0	0.068	0.035	0.0089	0.012	3.1	0.031[b]
	median	1.2	12	2.6	0.65	1.3	2.9	0.35	19	0.058	0.37	0.15	0.18	2.2	0.037	BDL	0.0044	0.0057	2.3	0.029[b]
	min	BDL	0.91	0.13	BDL	0.10	0.21	0.0020	1.4	BDL	BDL	0.0092	BDL	0.24	BDL	BDL	BDL	BDL	0.26	0.023[b]
	max	16	50	24	12	29	62	6.9	180	0.37	1.57	0.97	0.82	28	0.79	0.35	0.07	0.11	29	0.077[b]
	DF(%)	98	100	100	99	100	100	100	100	70	92	100	98	100	91	33	72	73	100	100[b]
urine ($n = 86$) ng/L	mean	2.4	16	5.5	0.62	3.9	10	0.93	37	1.9	2.2	0.46	0.42	79	0.73	0.45	0.28	0.34	81	0.88
	median	1.1	12	3.3	0.27	2.4	7.1	0.64	25	0.82	1.7	0.22	0.30	17	0.35	BDL	0.17	0.19	19	0.44
	min	BDL	0.72	0.45	BDL	BDL	0.66	BDL	2.0	BDL	BDL	BDL	BDL	2.3	BDL	BDL	BDL	BDL	2.6	BDL
	max	35	83	35	14	24	60.3	5.8	184	19	22.1	11	2.43	1033	4.1	7.1	2.7	4.1	1036	7.2
urine ($n = 86$) ng/g creatinine	mean	3.1	21	7.2	0.82	4.8	13	1.1	47	3.4	3.6	1.1	0.70	120	0.93	0.55	0.35	0.43	122	0.92
	median	1.4	14	4.0	0.28	2.5	7.5	0.70	28	1.0	1.8	0.27	0.36	19	0.34	0.18	0.20	0.27	23	0.61
	min	BDL	1.0	0.60	BDL	BDL	0.94	BDL	2.8	BDL	BDL	BDL	BDL	3.0	BDL	BDL	BDL	BDL	3.5	BDL
	max	51	109	47	20	20	59	5.8	232	73	84	42	9.2	1864	9.4	7.0	3.8	6.0	1869	5.5
	DF(%)	98	100	100	77	98	100	91	100	67	85	92	98	100	76	23	99	99	100	99

[a]To compute the descriptive statistics, values less than LOD have been replaced by LOD/$\sqrt{2}$. DF = detection frequency. Max = maximum; min = minimum; BDL = below detection limit. [b]Data about PFOSA were only from 30 whole blood samples and only linear PFOSA was quantificationally analyzed because of lack of branched PFOSA standards.

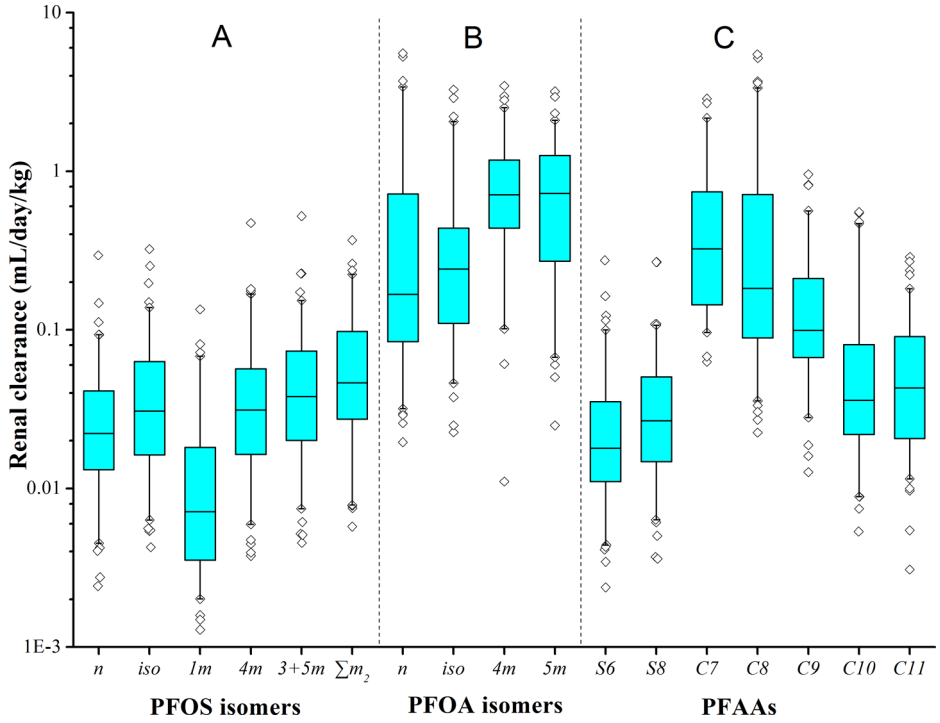

Figure 2. Distributions of estimated renal clearance for linear and branched (A) PFOS, (B) PFOA isomers, and (C) various chain-length PFAAs in human paired blood−urine samples. The lower and upper bounds of the boxes indicate the 25th and 75th percentiles, respectively. The horizontal lines within the boxes indicate median values. The lower and upper limits of the whiskers indicate 5% and 95% values, respectively, and circles below or above the whiskers indicate outlier values. S6 and S8 represent PFHxS and PFOS, respectively, while C7−C11 represent PFHpA−PFUnA.

Table 2. PFAA Renal Clearance (mL/day/kg)

	young female group					male and older female group				
compound	mean	95% CI[a]	GM	median	N[b]	mean	95% CI[a]	GM	median	N[b]
PFHxS	0.039	0.020−0.057	0.028	0.033	19	0.027	0.018−0.037	0.018	0.015	64
n-PFOS	0.045	0.032−0.057	0.037	0.038	20	0.031	0.021−0.042	0.020	0.019	66
iso-PFOS	0.062	0.040−0.083	0.047	0.047	20	0.045	0.032−0.059	0.028	0.027	66
1*m*-PFOS	0.019	0.0081−0.030	0.013	0.015	13	0.016	0.0092−0.023	0.0078	0.0061	53
4*m*-PFOS	0.057	0.041−0.074	0.048	0.050	20	0.049	0.032−0.067	0.027	0.028	64
3 + 5*m*-PFOS	0.072	0.051−0.094	0.062	0.069	20	0.053	0.035−0.071	0.032	0.033	66
$\sum m_2$-PFOS	0.093	0.052−0.13	0.064	0.073	16	0.063	0.048−0.078	0.044	0.044	62
∑PFOS	0.050	0.037−0.064	0.043	0.044	20	0.037	0.026−0.049	0.024	0.024	66
PFHpA	0.61	0.022−1.2	0.27	0.17	12	0.61	0.38−0.83	0.39	0.41	31
PFNA	0.25	0.13−0.37	0.14	0.20	16	0.15	0.099−0.20	0.10	0.094	50
PFDA	0.066	0.035−0.097	0.047	0.047	19	0.096	0.050−0.13	0.045	0.035	60
PFUnA	0.064	0.033−0.095	0.045	0.045	19	0.065	0.050−0.080	0.044	0.042	63
n-PFOA	0.29	0.094−0.48	0.15	0.14	20	0.79	0.48−1.1	0.26	0.18	66
iso-PFOA	0.57	0.13−1.0	0.30	0.26	15	0.39	0.22−0.56	0.21	0.22	47
4*m*-PFOA	0.78	0.40−1.2	0.59	0.52	12	0.92	0.70−1.1	0.61	0.76	49
5*m*-PFOA	1.2	0.67−1.7	0.83	1.0	15	0.75	0.57−0.92	0.50	0.64	48
∑PFOA	0.30	0.11−0.49	0.17	0.16	20	0.77	0.47−1.1	0.27	0.19	66
n-PFOSA	0.43	0.10−0.76	0.29	0.32	7	0.46	0.22−0.69	0.28	0.22	23

[a]CI = confidence interval. [b]Number of blood−urine pairs that were available for estimating renal clearance.

branched isomers detected in blood were 3+5*m*-PFOS (20%), *iso*-PFOS (12%), 4*m*-PFOS (9.3%), 1*m*-PFOS (4.4%), and $\sum m_2$-PFOS (1.4%). Note that 3*m*- and 5*m*-PFOS isomers could not be baseline separated and thus are reported as the sum. In urine samples, the mean percentage of *n*-PFOS was 45%, followed by 28% 3+5*m*-PFOS, 15% *iso*-PFOS, 9.8% 4*m*-PFOS, 2.5% $\sum m_2$-PFOS and 1.3% 1*m*-PFOS. Thus, the proportion of *n*-PFOS was significantly lower in urine than in blood samples, as was 1*m*-PFOS ($p < 0.001$ in both cases).

For PFOA, the dominant isomer in blood was *n*-PFOA (mean proportion 97%). Contrary to PFOS, but consistent with other literatures,[9,12,15] the proportion of *n*-PFOA in blood was distinctly higher than in the ECF standard (74%). In the urine samples, *n*-PFOA was still the predominant PFOA isomer (mean proportion 94%) but was less prominent than in blood (97%). Thus, for both PFOS and PFOA, the relative proportions of the linear isomer in urine were lower ($p <$

Table 3. Estimated Biological Elimination Half-Lives of PFAAs (Years)

	young females							all males and older females						
compound	mean	SE	GM	median	min	max	N^a	mean	SE	GM	median	min	max	N^a
PFHxS	7.7	0.6	7.1	7.1	2.3	13	19	35	3.9	25	29	1.6	182	64
n-PFOS	6.7	0.5	6.3	6.6	3.1	11	20	34	4.3	22	23	1.5	182	66
iso-PFOS	5.9	0.6	5.4	5.7	1.9	12	20	24	2.8	16	16	1.4	102	66
1*m*-PFOS	10	0.8	9.6	9.9	4.5	14	13	90	11	55	72	3.2	336	53
4*m*-PFOS	5.8	0.5	5.4	5.5	2.2	11	20	27	3.4	16	16	0.93	115	64
3 + 5*m*-PFOS	5.0	0.4	4.6	4.5	1.7	10	20	21	2.6	14	13	0.84	97	66
$\sum m_2$-PFOS	5.1	0.8	4.3	4.5	1.5	12	16	14	1.8	9.8	10	1.2	75	62
∑PFOS	6.2	0.5	5.8	6.0	3.2	10	20	27	3.1	18	18	1.6	121	66
PFHpA	1.5	0.3	1.0	1.6	0.11	3.3	12	1.2	0.2	0.82	0.79	0.12	5.1	31
PFNA	2.5	0.6	1.7	1.5	0.38	7.7	16	4.3	0.5	3.2	3.5	0.34	20	50
PFDA	4.5	0.4	4.0	4.2	1.2	7.7	19	12	1.5	7.1	9.2	0.58	60	60
PFUnA	4.5	0.5	4.0	4.4	1.1	7.8	19	12	2.0	7.4	7.7	1.1	104	63
n-PFOA	2.3	0.4	1.7	2.0	0.19	5.3	20	2.8	0.4	1.2	1.8	0.058	16	66
iso-PFOA	1.4	0.4	0.92	1.1	0.11	6.0	15	2.5	0.4	1.5	1.5	0.099	14	47
4*m*-PFOA	0.64	0.1	0.51	0.59	0.15	1.3	12	1.4	0.6	0.53	0.42	0.093	29	49
5*m*-PFOA	0.53	0.1	0.37	0.31	0.10	1.4	15	1.3	0.3	0.65	0.51	0.11	13	48
∑PFOA	2.1	0.3	1.5	1.8	0.19	5.2	20	2.6	0.4	1.2	1.7	0.059	14	66

[a]Number of blood−urine pairs that were available for estimating elimination half-lives of PFAAs.

0.001) than in paired blood samples, suggesting that branched isomers, in general, are preferentially excreted in urine.

Renal Clearance Estimation. We calculated the daily renal clearance (mL/day/kg) of individual PFAAs based on the paired serum and urinary concentrations. The calculated renal clearance using eq 1 was similar between volumetric and creatinine normalized urine data, with no significant differences for all PFAAs except for *iso*-PFOA (paired *t*-test). Thus, to simplify the discussion we only report the results based on urinary volume. Figure 2 compares the renal clearance estimates for all PFAAs, including the major isomers of PFOS and PFOA. PFHxS and PFOS showed much lower renal clearance efficiencies than PFHpA, PFOA, and PFNA (Figure 2C), suggesting that PFSAs are excreted through urine much less efficiently than PFCAs of the same perfluorinated carbon chain length. Among PFCAs, the renal clearance efficiency decreased as the perfluorinated carbon chain length increased up to C9−C10, and then remained constant for PFDA and PFUnA. In contrast to this structure−activity for PFCAs, the renal clearance efficiency of PFHxS (a C6 compound) was significantly lower than that of ∑PFOS (a C8 compound) (paired *t*-test, $p = 0.002$, $N = 83$). Among the major PFOS isomers (Figure 2A), 1*m*-PFOS had the lowest renal clearance efficiency, followed by $n < iso \approx 4m < 3{+}5m < \sum m_2$. The branched isomers of PFOA were all cleared more efficiently than *n*-PFOA, with the rank order being $n < iso < 4m \approx 5m$ (Figure 2B).

The relative proportions of linear and total branched PFOSA isomers in 30 paired samples of whole blood and urine were estimated based on chromatographic peak areas. The proportion of *n*-PFOSA was significantly higher in urine (median 84%) than in blood (65%) ($p < 0.001$) (see SI Figure S3), demonstrating that *n*-PFOSA is preferentially excreted via urine relative to branched isomers; opposite to the findings for urinary excretion of PFOS and PFOA isomers.

Elimination Half-Lives of PFAAs. Based on the literature discussed above, and strong associations between urinary and blood concentrations (except PFUnA), renal clearance was assumed to be the major pathway for human elimination of the PFCAs and PFSAs, thus, CL_{total} was set equal to CL_{renal} (mL/day/kg) in eq 2. However, considering that menstrual clearance is an important clearance pathway for PFAAs in young females,[28] menstrual clearance was estimated and added to renal clearance for calculation of CL_{total} in young females, using the same rate estimated by Harada et al.[28] for Japanese women (0.029 mL/day/kg). This estimated menstrual clearance is lower than CL_{renal} for most PFAAs (Table 2) and is a reasonable estimate because it is in the range of CL_{renal} for PFHxS and PFOS, which are two PFAAs that were significantly lower in younger females. Although other clearance mechanisms are acknowledged, as discussed above, they are believed to be minor and were not accounted for experimentally (albeit, fecal elimination might be important for some substances, see Discussion). Nevertheless, biological half-life estimates, which ranged from 0.5 to 10 years in young females, and from 1.2 to 90 years in males and older females (Table 3), should be considered as upper limit estimates of the biological half-life.

■ DISCUSSION

We observed significant correlations between PFAA concentrations in paired blood (ng PFAA/mL) and urine (ng PFAA/L), except for PFUnA, which is notably the longest chain length PFCA examined (a C11 compound). Beesoon et al. observed long chain-length PFCAs (C13 and C14) in human stool, but not in urine and serum.[26] Thus it is likely that fecal excretion becomes more important than urinary excretion for PFCAs longer than C10. Nevertheless, for the major PFAAs it is apparent that urine is an alternative matrix for biomonitoring, and can be considered in certain populations as it is less invasive and easier to collect than serum samples. Human nails were formerly suggested as a less invasive sample for human biomonitoring,[30] but this approach does not seem feasible for PFHxS and PFOA, which are not detectable in nails or do not correlate with blood.

Harada et al.[28] previously studied the renal clearance of PFOA and PFOS in 20 Japanese people, including 5 young females and 15 males and old females, using paired serum and 24-h pooled urine samples. The median renal clearance efficiencies of PFOS in younger females, males and older females were estimated to be 0.014 and 0.016 mL/day/kg,

respectively, which were lower than those in the current work (0.044 and 0.024 mL/day/kg, respectively). The median renal clearance efficiencies of PFOA were 0.046 and 0.022 mL/day/kg in younger females, males and older females,[28] which were much lower than those in the present study (0.16 and 0.19 mL/day/kg, respectively). The reason for these differences is unclear (race and sampling size may be a factor), but in both cases the consistent finding was that PFOS is excreted more slowly through urine than PFOA. It is notable that the estimated PFOA elimination half-life (~ 2−3 years, Table 3) in the current study, which was derived from our clearance estimates, is similar to the actual PFOA serum elimination half-lives reported for highly exposed populations (i.e., 2.3 or 2.9 years).[10,11]

To our knowledge, this is the first study investigating the excretion of long-chain PFCAs via urine in humans. It was demonstrated that PFCAs with increasingly longer carbon chain lengths (from C7 to C10) were generally more difficult to excrete than shorter ones (Figure 2C). This is a sensible result, as shorter PFCAs are more water-soluble than longer ones.[43] In addition, the affinities of PFAAs to organic anion transport proteins could also affect their excretion. It was reported that PFCAs with longer carbon chain length displayed stronger interaction with organic anion transport polypeptide (OATP) 1a1, leading to lower elimination of these compounds with longer carbon chain length (from C6 to C10).[44] Previous studies of PFCAs (from PFHpA to PFDA) have shown that urinary elimination from rats also becomes less efficient as the perfluorocarbon chain length increases.[25,39] What could not be explained by physical properties, however, was that PFHxS was more difficult to excrete in urine than PFOS, even though it has a shorter carbon chain length than PFOS. Nevertheless, the current data are consistent with those of Olsen et al., who reported a longer serum elimination half-life for PFHxS (8.5 years) than PFOS (5.4 years) in retired occupational workers.[9]

The menstrual serum clearance rate (0.029 mL/day/kg) was lower than the renal clearance rate of most PFAAs but was comparable to the renal clearance estimated for PFHxS (0.033 mg/day/kg) and PFOS (0.044 mL/day/kg) and longer carbon chain PFCAs, such as PFDA (0.047 mL/day/kg) and PFUnA (0.045 mL/day/kg) (Table 2). Menstruation may therefore be an important excretion route of PFAAs in young females, but only for those substances that are most difficult to eliminate through urine.

The estimated geometric mean elimination half-lives of PFHxS and PFOS for the young female group were 7.1 and 5.8 years, respectively, while they were 25 and 18 years, respectively, for males and older females (Table 3). It has been reported that the geometric mean elimination half-lives of PFHxS and PFOS were 7.3 and 4.8 years, respectively, for older (age ≥55 years) occupational workers, including 24 males and 2 females.[9] As discussed, owing to the possibility that excretion pathways other than urine may be significant, the estimated elimination half-lives reported in the current study can be viewed as upper limits. The relatively lower blood concentrations of PFHxS, PFOS, and PFNA in the young female group can be partly explained by their relatively shorter half-lives in this group.

For PFOA the estimated arithmetic mean half-lives were 2.1 and 2.6 years in the young female, and male and older female groups, respectively (Table 3). These data are comparable with previously reported results of 3.8 years,[9] and 2.3 or 2.9 years for people with high exposure.[10,11] Thus, for PFOA it is likely that renal excretion is the dominant elimination mechanism, but for PFHxS and PFOS, and longer chain PFCAs (i.e., PFDA and PFUnA), other routes of elimination play a role and need to be examined quantitatively in future studies. More data on the volume of distribution for all analytes examined would also improve the current estimates of biological half-life.

In general, the branched isomers of PFOS and PFOA were preferentially excreted in urine compared with *n*-PFOS and *n*-PFOA (Figure 2A and B). The remarkable exception was that the renal clearance efficiency of 1*m*-PFOS was lower than *n*-PFOS, and was the lowest among all PFOS isomers. In fact, its urinary elimination was even lower ($p < 0.001$) than PFHxS and was the most persistent of all PFAA compounds examined (Figure 2). These data are consistent with rodent data,[16] whereby branched PFOS and PFOA were also preferentially excreted, with the exception of 1*m*-PFOS which was also the most persistent PFOS isomer.

Assuming that renal elimination is the primary pathway of PFOA and PFOS excretion, and that ECF PFOA (~80% linear) and PFOS (~70% linear) are important sources of human exposure, then the faster renal excretion of the major branched isomers should lead to enriched proportions of linear PFOA (i.e., >80%) and PFOS (i.e., >70%) in human blood. This was true for PFOA in the current study, and is consistent with other studies.[12,15,32] However, the proportion of *n*-PFOS in Chinese human blood was significantly lower (i.e., mean = 53% linear) than historical ECF standards, similar to previous monitoring efforts in other countries.[13] This consistent finding cannot be explained by the PFOS isomer pharmacokinetic parameters reported in the current work. As previously discussed and proven in vitro,[45] one plausible alternative explanation for the enriched branched PFOS content is that the metabolism of branched PFOS precursors to branched PFOS is more rapid and efficient than the metabolism of linear PFOS precursors to linear PFOS. Data from the current work raises an additional mechanism whereby *n*-PFOSA was excreted in urine more rapidly than branched PFOSA. Thus, more branched PFOSA are bioavailable to be metabolized to branched PFOS. This is an important observation because a wide variety of PFOS precursors are metabolized to PFOS via PFOSA as the common intermediate metabolite.[46] It is likely that the combination of preferential excretion of linear PFOS precursors, and the faster metabolism of branched PFOS precursors to PFOS, act cumulatively to produce an enriched branched PFOS profile in human serum. Thus, enriched branched PFOS profiles in humans may indeed be a good biomarker of human exposure to PFOS precursors.

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

Detailed information on the urinary creatinine analysis method, HPLC-MS/MS analysis method, chromatogram of isomers, acronyms, summed demographic information about the 86 participants, LOD and recoveries for PFAAs and PFOSA, relationship between the concentrations of each PFAA expressed with either units in urine, between the blood and paired urine concentrations, figures describing the levels of PFAAs in the four categories and *n*-PFOSA proportion with peak area in the 30 paired whole blood−urine samples. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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